

Qatar-1b

R-2758

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_140819 · created 2026-09-03T00:08:25+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456888.73471 +/- 0.00035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.056
Transit depth (Rp/Rs)^2	3.13 +/- 1.99 [%]
Orbital Inclination (inc)	83.13 +/- 2.82
Airmass coefficient 1 (a1)	1.9 +/- 0.19
Airmass coefficient 2 (a2)	-0.49 +/- 0.3
Scatter in the residuals of the lightcurve fit is	10.16 %
Best Comparison Star	#1 - [315, 165]
Optimal Aperture	1.6
Optimal Annulus	4.91
Transit Duration (day)	0.055 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.177 $\pm$ 0.056 vs 0.1463 (deviation 0.43 σ)
Duration fit vs published	0.055 d vs 0.0692 d (deviation 0.53 σ)
Mid-time O-C	0.5 min
Depth SNR	2.5
Residual scatter	10.16 %

Light curve

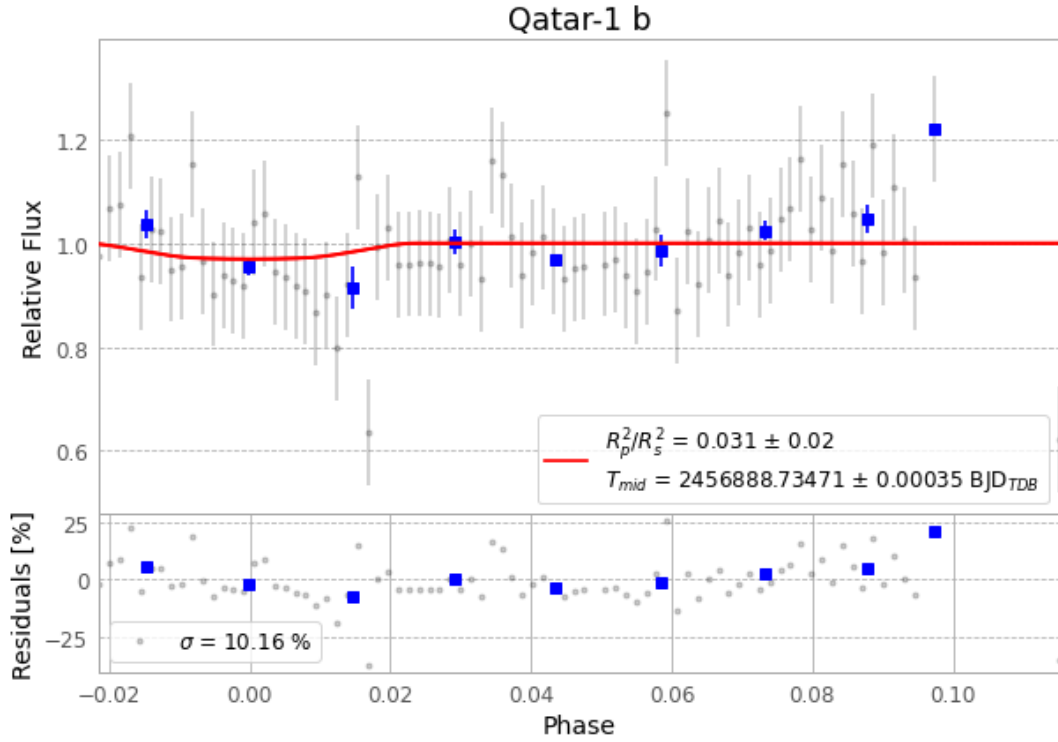


Figure 1 — FinalLightCurve_Qatar-1 b_2014-08-18.png (SHA-256 a740ffe088ea298c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [337, 318], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 838b4ea7479b6092...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

98 FITS frames (64 MB), Qatar-1140819045112.FITS → Qatar-1140819094213.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-140819.log (3d409b54c2e9...), FinalLightCurve_Qatar-1 b_2014-08-18.png (a740ffe088ea...), FinalLightCurve_Qatar-1 b_2014-08-18.csv (a776f3671922...)

Compute resources

Wall time 414.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 00:08Z.